How strongly do trace-characterising inequalities fail? Sharp violation envelopes in terms of the Michelson contrast

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Abstract

Many inequalities that hold for the trace characterise it: a positive functional that satisfies one of them for all operators is tracial. Examples are Gardner's inequality $|\varphi(X)| \le \varphi(|X|)$, the triangle inequality for $\varphi(|\cdot|)$, monotonicity, Young's, Araki–Lieb–Thirring's and determinant inequalities, and a family of inequalities for pairs of projections. We ask how strongly such an inequality fails for a given non-tracial positive functional $\varphi = \mathrm{Tr}(W\,\cdot\,)$ on M_n, and measure this by the *violation envelope*: the largest normalised violation over all operators, as a function of the Michelson contrast δ of the density W. For 25 characterisations from the literature we determine the envelope exactly in sixteen cases (three of them only in dimension two or in even dimension), give explicit lower bounds in four, and compute the remaining ones by a certified numerical search. The envelopes fall into classes by the order in which they vanish at the trace: the extended Young inequality, two lattice inequalities for projections, an inequality of Bikchentaev and Abed and the power-mean inequality for the canonical trace fail linearly in δ; Gardner's inequality, its relatives and most inequalities for projections fail quadratically; the sharp constants are expressed through the Kantorovich constant, the condition number and the Specht ratio of W. Monotonicity fails at least as $\delta^4/16$ and its weighted form at least as $\delta^8/4096$. On a direct sum $\bigoplus_c M_{k_c}$ the envelope is governed by the contrasts inside the blocks alone: a density that varies only between blocks is invisible to every characterisation of traces. Finally, we determine the sharp upper bound of the unnormalised violation $|\mathrm{Tr}(XZ)| - \mathrm{Tr}(X|Z|)$ of Gardner's inequality when both contrasts $\Delta(X)$ and $\Delta(|Z|)$ are prescribed, on every direct sum of real or complex matrix algebras, which answers a question raised in earlier work by Abed and the authors.

Keywords: trace characterisation; positive functional; Michelson contrast; Kantorovich inequality; Specht ratio; two projections; von Neumann algebra.
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1 Introduction

Let φ be a positive linear functional on a C*-algebra. Gardner [18] proved that φ is a trace, $\varphi(XY) = \varphi(YX)$, if and only if

$$|\varphi(X)| \le \varphi(|X|) \qquad \text{for all } X. \tag{1}$$

This started a long series of results of the same form: an inequality that holds for every trace holds for all operators only if φ is a trace. On M_n or among the positive normal functionals on a von Neumann algebra, traces are characterised by the triangle inequality for $\varphi(|\cdot|)$ [18, 30], by $|\varphi(A^k)| \le \varphi(|A|^k)$ [27], by monotonicity $0 \le A \le B \Rightarrow \varphi(A^p) \le \varphi(B^p)$ [5, 28] and its weighted forms [21, 22], by Young's inequality [4, 16], by subadditivity of $\varphi \circ f$ for concave f [31], by the Peierls–Bogoliubov and Araki–Lieb–Thirring inequalities [9], by the generalised Powers–Størmer inequality [23], by inequalities for determinants and power means [12], by the spectral geometric mean [15], by inequalities for $|\varphi(A^2)|$ [14], by metrics on projections [11], by subadditivity on the projection lattice [27, Theorem 4], and by a family of inequalities in which only projections enter [6, 7, 8, 10]. Bikchentaev and Moslehian [13]

survey these results; the canonical trace on M_n is characterised in [25, 12]. The dual problem, in which the functional is arbitrary and the operator is to be characterised as central, is treated in [26].

All these theorems answer a yes/no question. Here we ask a quantitative one: *how strongly does a trace-characterising inequality fail for a given non-tracial functional?* On M_n every positive functional has the form $\varphi_W = \mathrm{Tr}(W\,\cdot\,)$ with a density $W \geq 0$, and φ_W is a trace if and only if W is a scalar multiple of 1. The natural measure of the distance of W from the scalars that is invariant under scaling is the Michelson contrast

$$\Delta(W) = \frac{\lambda_{\max}(W) - \lambda_{\min}(W)}{\lambda_{\max}(W) + \lambda_{\min}(W)} \in [0, 1],$$

studied for operators in [1]. There, a Monte Carlo study [1, Sect. 5.1, Fig. 1] plotted the violation of Gardner's inequality against $\Delta(W)$ for real 2×2 and 3×3 matrices and conjectured that such violations are bounded by functions of the contrasts [1, Sect. 7]. We make this precise. For an inequality $L \leq R$ (with L, R depending on φ and on operator variables) the *violation envelope* is

$$V_n(\delta) = \sup\Big\{ \frac{L - R}{N} : \ W \geq 0 \text{ in } M_n,\ \Delta(W) = \delta,\ \text{all operator variables free} \Big\},$$

where $N > 0$ is a normalisation that makes the quotient invariant under the scaling of W and of the variables (mostly $N = R$). By the characterisation theorems, $V_n(\delta) > 0$ for $\delta > 0$; by construction $V_n(0) = 0$.

Results. We study the 25 inequalities of Table 1. Write $a \geq b > 0$ for the extreme eigenvalues of W and

$$h = \frac{a}{b} = \frac{1+\delta}{1-\delta}, \qquad \kappa = \frac{a+b}{2\sqrt{ab}} = \frac{1}{\sqrt{1-\delta^2}}, \qquad \psi = \arcsin\delta,$$

the condition number, the Kantorovich constant and the contrast angle of W.

Theorem A. *The envelopes of the inequalities of Table 1 are as follows.*

(a) $V_n = \kappa - 1$ *for Gardner's inequality, the triangle inequalities for* $\varphi(|\,\cdot\,|)$ *on operators and on projections,* $\varphi(A+A^*) \leq 2\varphi(|A^*|)$, $|\varphi(A^2)| \leq \varphi(A^*A)$, $\varphi(|P-Q|) \leq \varphi(P+Q)$ *and* $\varphi(\mathrm{Re}\,PQ) \leq \varphi(|PQ|)$, *in every dimension* $n \geq 2$ *(Theorem 3.1, Theorem 5.1).*

(b) $V_n = \kappa^2 - 1$ *for* $\varphi(PQP) \leq \varphi(Q)$; $V_n = \frac{1}{2}\tan^2(\psi/2)$ *for* $\varphi(\mathrm{Re}\,PQ) \geq 0$; $V_n = \delta/(1-\delta)$ *for* $\varphi(P \vee Q) \leq \varphi(P) + \varphi(Q)$; $V_n = \delta$ *for* $\varphi(P+Q) \leq \varphi(1 + P \wedge Q)$ *(n even); and* V_n *is the maximum of an explicit function of one variable,* $\frac{(e-1)^2}{e(e+1)}\delta^2 + O(\delta^3)$, *for* $\varphi(e^{QPQ}) \leq \varphi(e^P)$ *(n even) (Theorem 5.1).*

(c) $V_n = h - 1$ *for the extension of Young's inequality to arbitrary operators;* $V_2 = s/(1+s)$, $s = \sin(\psi/2)$, *for* $\varphi(PQ+QP) \leq 2\varphi(QPQ)$; *for the inequalities that characterise the* canonical *trace,* $V_2 = 1 - \sqrt{1-\delta^2}$ *for the AM–GM inequality* $(\det A)^{1/n} \leq \frac{1}{n}\varphi(A)$, *increasing with n to* $1 - 1/S(h)$ *with the Specht ratio S, and* $V_n = \sqrt{(\ell + n - 1)/(n\ell)} - 1$, $\ell = 1/h$, *for* $\varphi(A^2)^{1/2} \leq \varphi(A)$ *(Section 4).*

(d) $V_n \geq \kappa - 1$ *for the spectral geometric mean inequality,* $V_n \geq \kappa^2 - 1$ *for the Araki–Lieb–Thirring inequality,* $V_n \geq \delta^4/16 + O(\delta^5)$ *for monotonicity and* $V_n \geq \delta^8/4096\,(1 + O(\delta^2))$ *for its weighted form with* $r = 1$; *and* $V_n \leq h - 1$ *for the first ten items (Section 6).*

Thus the envelopes vanish at the trace to different orders, δ, δ^2, δ^4 and δ^8 (Table 2): some characterisations are far more sensitive than others, and two inequalities that are equivalent for traces may fail at different rates. For instance, $\varphi(P\vee Q)+\varphi(P\wedge Q)=\varphi(P)+\varphi(Q)$ for every trace, yet the lattice subadditivity fails as $\delta/(1-\delta)$ and $\varphi(|P-Q|)\le\varphi(P+Q)$, with the same right-hand side, only as $\kappa-1\sim\delta^2/2$.

The second result concerns finite-dimensional von Neumann algebras $\mathcal{M}=\bigoplus_{c=1}^{C}M_{k_c}$. A positive functional φ_W on $\mathcal{M}$ is a trace if and only if W is central, i.e. scalar on every block, however different the scalars are. The violation therefore cannot depend on the global contrast of W:

Theorem B. *Let $W=\bigoplus_c W_c>0$ in $\mathcal{M}$. For every inequality of Table 1 except 11, 16 and 24, the largest normalised violation for φ_W on $\mathcal{M}$ is at most the largest one for φ_{W_c} on the blocks M_{k_c}, with equality for items 1–10, 12–14, 17, 18, 21–23 and 25. In particular a density that varies only between the blocks produces no violation at all (Theorem 7.1).*

The third result returns to the unnormalised quantity of [1], in which both operators are normalised to norm one and both contrasts are prescribed:

Theorem C. *Let $\mathcal{M}=\bigoplus_{c=1}^{C}M_{k_c}(\mathbb{K})$, $\mathbb{K}\in\{\mathbb{R},\mathbb{C}\}$, $N=\sum_c\lfloor k_c/2\rfloor\ge 1$. For $X\ge 0$ and Z in $\mathcal{M}$ with $\|X\|=\|Z\|=1$, $\Delta(X)=\delta<1$ and $\Delta(|Z|)=\delta'$,*

$$\sup\bigl(|\operatorname{Tr}(XZ)|-\operatorname{Tr}(X|Z|)\bigr)=\frac{N\,\delta^2\delta'^2}{(1+\delta)(1+\delta')},$$

and the supremum is attained (Theorem 8.2).

All values in Theorems A and C are confirmed by a certified numerical search over all variables, which also gives the envelopes of the five inequalities without a closed form (Section 9).

What is new. The characterisation theorems are due to the cited authors, and the constants κ, h and the Specht ratio, as well as the inequalities behind them (Kantorovich's inequality in the form of Greub and Rheinboldt [19, 17], Specht's reverse AM–GM inequality [29]), are classical. The upper bounds of Theorem 3.1 are the Kantorovich inequality; what we add is that the resulting constants are the exact envelopes. We found no previous study of the envelopes; the closest work is [1], which plots the unnormalised Gardner violation, defines no envelope and proves no bound. A short account of part of these results appears as a use case in the description of the software used for the computations [24].

Organisation. Section 2 fixes the notation, lists the inequalities and collects the tools. Sections 3–6 prove Theorem A, Section 7 Theorem B and Section 8 Theorem C. Section 9 describes the numerical verification and the remaining items, and Section 10 lists open problems.

2 Setting and tools

2.1 Functionals, contrast and envelopes

Throughout, Tr is the usual trace on $M_n=M_n(\mathbb{C})$, $|X|=(X^*X)^{1/2}$, $\|X\|_1=\operatorname{Tr}|X|$, $\operatorname{Re}X=\frac12(X+X^*)$, and $\varphi=\varphi_W=\operatorname{Tr}(W\,\cdot\,)$ with $W\ge 0$; φ is a trace iff $W\in\mathbb{C}1$. We write $a\ge b$ for the extreme eigenvalues of W, e_1,e_2 for corresponding unit eigenvectors, and $\delta=\Delta(W)$; unless stated otherwise $W>0$, i.e. $\delta<1$. The quotient $(L-R)/N$ is invariant under $W\mapsto tW$, $t>0$, and the envelope $V_n(\delta)$ is the supremum over all W with $\Delta(W)=\delta$; for a fixed density we write $V(W)$. We call a value of an envelope *attained* if the supremum is a maximum and *approached* otherwise. Variables called P,Q,R are orthogonal projections, $A,B\ge 0$ unless said otherwise, and X,Y arbitrary; $P\wedge Q$ and $P\vee Q$ are the projections onto $\operatorname{ran}P\cap\operatorname{ran}Q$ and onto $\operatorname{ran}P+\operatorname{ran}Q$.

	inequality $L \le R$	variables	N	source
1	$\lvert\varphi(X)\rvert \le \varphi(\lvert X\rvert)$	X	R	[18]
2	$\varphi(\lvert X+Y\rvert) \le \varphi(\lvert X\rvert)+\varphi(\lvert Y\rvert)$	X, Y	R	[18, 30]
3	$\varphi(A+A^*) \le 2\varphi(\lvert A^*\rvert)$	A arbitrary	R	[10]
4	$\lvert\varphi(A^2)\rvert \le \varphi(A^*A)$	A arbitrary	R	[27, Thm. 1, $k=2$], [14]
5	$\varphi(\lvert P-R\rvert) \le \varphi(\lvert P-Q\rvert)+\varphi(\lvert Q-R\rvert)$	P, Q, R	R	[11]
6	$\varphi(A \natural B) \le \sqrt{\varphi(A)\varphi(B)}$	$A, B > 0$	R	[15]
7	$\varphi((B^{1/2}AB^{1/2})^2) \le \varphi(BA^2B)$	A, B	R	[9, Lemma 2, $r=2$, $p=1$]
8	$\varphi(\lvert AB^*\rvert) \le \frac{1}{2}\varphi(\lvert A\rvert^2)+\frac{1}{2}\varphi(\lvert B\rvert^2)$	A, B arbitrary	R	extension of [4]
9	$\varphi(PQ+QP) \le 2\varphi(QPQ)$	P, Q	$\varphi(P)+\varphi(Q)$	[10]
10	$0 \le A \le B \Rightarrow \varphi(A^2) \le \varphi(B^2)$	$A \le B$	R	[5, 28]
11	$(\det A)^{1/n} \le \frac{1}{n}\varphi(A)$, $\varphi(1)=n$	A	$(\det A)^{1/n}$	[12, Thm. 3(iii)]
12	$\varphi((A+B)^{1/2}) \le \varphi(A^{1/2})+\varphi(B^{1/2})$	A, B	R	[31, Cor. 1]
13	$\varphi(A+B-\lvert A-B\rvert) \le 2\varphi(A^{1/4}B^{1/2}A^{1/4})$	A, B	$\varphi(A+B)$	[23, Lemma 3.1]
14	$\lvert\varphi(PQ)\rvert \le \varphi(P)^{1/3}\varphi(Q)^{2/3}$	P, Q	R	[8, Thm. 2(i)]
15	$\varphi(e^{P+Q}) \le \varphi(e^{P/2}e^Qe^{P/2})$	P, Q	R	[8, Thm. 2(iv)]
16	$\varphi(e^A)\exp\frac{\varphi(e^{A/2}Be^{A/2})}{\varphi(e^A)} \le \varphi(e^{A+B})$	A, B self-adj.	R	[9, Lemma 1]
17	$\varphi(PQP) \le \varphi(Q)$	P, Q	R	[7, Thm. 3.4(iv)]
18	$\varphi(\lvert P-Q\rvert) \le \varphi(P+Q)$	P, Q	R	[7, Thm. 3.4(v)]
19	$\varphi(P+Q) \le \varphi(1+P\wedge Q)$	P, Q	R	[7, Thm. 3.4(iii)]
20	$\varphi(e^{QPQ}) \le \varphi(e^P)$	P, Q	R	[8, Thm. 2(v)]
21	$0 \le \varphi(\operatorname{Re} PQ)$	P, Q	$\varphi(P)+\varphi(Q)$	[8, Thm. 2(vi)]
22	$\varphi(\operatorname{Re} PQ) \le \varphi(\lvert PQ\rvert)$	P, Q	R	[8, Thm. 2(vii)]
23	$\varphi(P\vee Q) \le \varphi(P)+\varphi(Q)$	P, Q	R	[27, Thm. 4]
24	$\varphi(A^2)^{1/2} \le \varphi(A)$, $\varphi(1)=n$	A	R	[12, Thm. 3(vii)]
25	$0 \le A \le B \Rightarrow \varphi(A^4) \le \varphi(AB^2A)$	$A \le B$	R	[22, Thms. 4, 5, $r=1$], [21]

Table 1: The inequalities studied, with the normalisation N of the violation $(L-R)/N$ and the source of the characterisation. Projections are P, Q, R; positive operators A, B unless stated otherwise. Remarks on the forms are in Section 2.2.

2.2 Remarks on the forms

Several sources state families of inequalities; we use one member. In item 4, $\lvert\varphi(A^k)\rvert \le \varphi(\lvert A\rvert^k)$ for all A and one fixed $k \ge 1$ characterises the trace on M_n [27, Theorem 1]; $k=1$ is Gardner's inequality. Item 7 is the Araki–Lieb–Thirring inequality $\varphi((B^{1/2}AB^{1/2})^{rp}) \le \varphi((B^{r/2}A^rB^{r/2})^p)$ with $r=2$, $p=1$ [9, Lemma 2]. The published form of item 8 is $\varphi(\lvert AB\rvert) \le \frac{1}{p}\varphi(A^p)+\frac{1}{q}\varphi(B^q)$ for $A, B \ge 0$ [4]; we use $p=q=2$ and allow arbitrary A, B, in the form of Ando's matrix Young inequality [2], and Proposition 4.1 refers to this extension. Item 9 is the member $p=1$ of the family $\varphi(PQ+QP) \le 2\varphi((QPQ)^p)$, $p \in (0,1]$ [10]. Items 10 and 25 are the cases $r=0$ and $r=1$ of $\varphi(A^{2r+2}) \le \varphi(A^rB^2A^r)$ for $0 \le A \le B$, which characterises traces for every fixed $r \ge 0$ [22, Lemma 1, Theorems 4, 5]; on M_n the case $r=0$ is in [5, 28] and $r>0$ in [21, Theorem 6]. Items 11 and 24 characterise the canonical trace among the positive functionals with $\varphi(1)=n$ [12, Theorem 3]; item 24 is (vii) there with $p=2$, $q=1$ and needs φ faithful. Item 12 is the case $f(t)=t^{1/2}$ of Tikhonov's characterisation by $\varphi(f(A+B)) \le \varphi(f(A))+\varphi(f(B))$ for concave f with $f(0)=0$ [31, Corollary 1]. Item 13 is the generalised Powers–Størmer inequality $\varphi(A+B)-\varphi(\lvert A-B\rvert) \le 2\varphi(f(A)^{1/2}g(B)f(A)^{1/2})$, $g(t)=t/f(t)$, with $f(t)=t^{1/2}$ [23, Lemma 3.1, Remark 4]; with $N=R$ its envelope would be infinite (Remark 6.3), so we normalise by $\varphi(A+B)$. Item 14 is Hölder's inequality for projections with the exponent $\alpha=3$; for $\alpha=2$ it holds for every positive φ [8, remark after Theorem 2]. Item 16 is the Peierls–Bogoliubov inequality, stated in [9] for positive H, K; both sides scale by e^c under $A \mapsto A+c$,

$B \mapsto B+c$, so the envelope over self-adjoint arguments is the same. Items 17–19 are [7, Theorem 3.4], announced in [6], and items 14, 15, 20–22 are [8, Theorem 2].

2.3 Tools

Lemma 2.1 (Kantorovich inequality [19]). *If $b \le W \le a$, $a \ge b > 0$, then $\omega(W^2)\,\omega(1) \le \kappa^2\,\omega(W)^2$ for every positive functional ω, where $\kappa = (a+b)/(2\sqrt{ab})$.*

Proof. $(a-W)(W-b) \ge 0$ gives $\omega(W^2) \le (a+b)x - ab\,y$ with $x = \omega(W)$, $y = \omega(1)$, and $((a+b)x - aby)y \le \kappa^2 x^2$ because the difference is $(\kappa x - \sqrt{ab}\,y)^2$. □

Lemma 2.2 (Two projections; Halmos [20]). *For projections P, Q on a finite-dimensional Hilbert space H, H is the orthogonal sum of subspaces of dimension 1 or 2 invariant under P and Q. On each two-dimensional one there is an orthonormal basis f, f' with $P = ff^*$ and $Q = qq^*$, $q = \cos\theta\, f + e^{i\chi}\sin\theta\, f'$, $0 < \theta < \pi/2$.*

The next lemma gives a general upper bound; it shows that no envelope of items 1–8 and 10 can exceed the order δ.

Lemma 2.3. *For items 1–8 and 10, $V(W) \le h - 1$.*

Proof. In each item $L \le aL_0$, where L_0 is the expression for the trace or the trace norm (e.g. $|\varphi(X)| \le a\|X\|_1$, $\varphi(|X+Y|) \le a\,\mathrm{Tr}\,|X+Y|$, $|\varphi(A^2)| \le a\|A^2\|_1$), $L_0 \le R_{\mathrm{Tr}}$ by the trace inequality (for item 4 $\|A^2\|_1 \le \|A\|_2^2$; for item 7 the Araki–Lieb–Thirring inequality; for item 6, $A\natural B$ is unitarily similar to $(A^{1/2}BA^{1/2})^{1/2}$, so $\mathrm{Tr}(A\natural B) = \|A^{1/2}B^{1/2}\|_1 \le (\mathrm{Tr}\,A\,\mathrm{Tr}\,B)^{1/2}$), and $R \ge bR_{\mathrm{Tr}}$. Hence $L/R \le a/b = h$. For item 8, $\varphi(|AB^*|) \le a\|A\|_2\|B\|_2 \le \frac{a}{2}(\|A\|_2^2 + \|B\|_2^2)$. □

3 The Kantorovich class

Theorem 3.1. *Let $W > 0$ in M_n, $n \ge 2$, and $\varphi = \varphi_W$. For all X, Y, A and all projections P, Q, R,*

$$|\varphi(X)| \le \kappa\,\varphi(|X|), \qquad \varphi(A+A^*) \le 2\kappa\,\varphi(|A^*|), \qquad |\varphi(A^2)| \le \kappa\,\varphi(A^*A),$$
$$\varphi(|X+Y|) \le \kappa\big(\varphi(|X|) + \varphi(|Y|)\big), \qquad \varphi(|P-R|) \le \kappa\big(\varphi(|P-Q|) + \varphi(|Q-R|)\big).$$

All five constants are attained. Hence $V_n = \kappa - 1$ for items 1–5 in every dimension $n \ge 2$.

Proof. Let $X = U|X|$ be a polar decomposition and $T = |X|$. Then $\varphi(X) = \mathrm{Tr}(WUT) = \mathrm{Tr}\big((WT^{1/2})^*UT^{1/2}\big)$, so for every contraction U

$$|\varphi(X)|^2 \le \mathrm{Tr}(W^2T)\ \mathrm{Tr}\,T \le \kappa^2(\mathrm{Tr}\,WT)^2$$

by the Cauchy–Schwarz inequality and Lemma 2.1 for $\omega = \mathrm{Tr}(T\,\cdot\,)$. The second inequality follows in the same way from $\varphi(A+A^*) = 2\mathrm{Re}\,\varphi(|A^*|V)$, where $A = |A^*|V$ is a polar decomposition. For the fourth, write $X+Y = S|X+Y|$; then $\varphi(|X+Y|) = \varphi(S^*X) + \varphi(S^*Y)$, and each term is bounded as above with the contraction S^*U_X or S^*U_Y in place of U. The fifth is the fourth with $X = P-Q$, $Y = Q-R$. For the third, put $X = AW^{1/2}$ in an eigenbasis of W with eigenvalues w_i. Then $\varphi(A^*A) = \sum_{ij}|X_{ij}|^2$ and

$$\varphi(A^2) = \sum_i X_{ii}^2 + \sum_{i<j} X_{ij}X_{ji}\frac{w_i+w_j}{\sqrt{w_iw_j}}, \qquad \frac{w_i+w_j}{\sqrt{w_iw_j}} \le 2\kappa,$$

so $|\varphi(A^2)| \le \sum_i |X_{ii}|^2 + \kappa\sum_{i<j}(|X_{ij}|^2 + |X_{ji}|^2) \le \kappa\varphi(A^*A)$.

Equality, with e_1, e_2 eigenvectors of W for a, b: in the first two inequalities for $T = vv^*$ with $|\langle e_1, v\rangle|^2 = \frac{b}{a+b}$, $|\langle e_2, v\rangle|^2 = \frac{a}{a+b}$ and $Uv = Wv/\|Wv\|$ (equality in Cauchy–Schwarz and in Lemma 2.1); in the third for $A = \sqrt{a}\, e_1 e_2^* + \sqrt{b}\, e_2 e_1^*$; in the fourth for $X = uu^*$, $Y = -u'u'^*$ with $u, u' = \sin t\, e_1 \pm \cos t\, e_2$ and $\tan t = \sqrt{b/a}$; in the fifth for $P = uu^*$, $Q = 0$, $R = u'u'^*$. □

Remark 3.2. For a fixed density, $\sup_X |\varphi(X)|/\varphi(|X|) = \kappa(W)$ exactly. In particular the normalised violation of Gardner's inequality depends on W only through its extreme eigenvalues, not on the dimension or on the rest of the spectrum.

4 Young, Bikchentaev–Abed and the canonical trace

Proposition 4.1. *$\varphi_W(|AB^*|) \le h \cdot \frac{1}{2}\big(\varphi_W(|A|^2) + \varphi_W(|B|^2)\big)$ for all A, B, with equality for $A = B = e_1 e_2^*$. Hence $V_n = h - 1 = 2\delta/(1-\delta)$ for item 8 in every dimension.*

Proof. $\varphi_W(|AB^*|) \le a\|AB^*\|_1 \le a\|A\|_2\|B\|_2$ and $\varphi_W(|A|^2) = \mathrm{Tr}(WA^*A) \ge b\|A\|_2^2$. For $A = B = e_1 e_2^*$, $|AB^*| = e_1 e_1^*$ and $|A|^2 = e_2 e_2^*$, so $L = a$ and $R = b$. □

Remark 4.2. For the published form with $A, B \ge 0$ an independent numerical search ($n = 2, 3$) gives only $V \approx \kappa - 1$ (0.1546 at $\delta = 0.5$ and 1.294 at $\delta = 0.9$, against $\kappa - 1 = 0.1547$ and 1.294). The linear failure of item 8 is a feature of the extension to arbitrary operators.

Proposition 4.3. *For $n = 2$, item 9 has the envelope $V_2 = s/(1+s)$, $s = \sin(\psi/2)$. Thus $V_2 \sim \delta/2$ as $\delta \to 0$ and $V_2 \to \sqrt{2} - 1$ as $\delta \to 1$.*

Proof. If P or Q is 0 or 1, or $P = Q$, then $L = R$. Otherwise $P = uu^*$, $Q = vv^*$ with unit vectors; choose phases so that $u = cv + \sigma v_\perp$ with $c = \cos\theta$, $\sigma = \sin\theta$, $0 < \theta < \pi/2$, and a unit vector $v_\perp \perp v$. Normalise $\mathrm{Tr}\, W = 2$, so that W has the eigenvalues $1 \pm \delta$, and put $\beta = v^*Wv$, $\gamma = v_\perp^* W v_\perp = 2 - \beta$, $t = \mathrm{Re}\,(v^*Wv_\perp)$ and $\rho = |v^*Wv_\perp|$; then $\beta\gamma - \rho^2 = \det W = 1 - \delta^2$. Since $PQ + QP = c(uv^* + vu^*)$ and $QPQ = c^2 Q$,

$$L - R = 2c\mathrm{Re}\,(v^*Wu) - 2c^2\beta = 2c\sigma t, \qquad N = u^*Wu + \beta = \beta(1+c^2) + \sigma^2\gamma + 2c\sigma t.$$

With $D = \beta(1+c^2) + \sigma^2\gamma > 0$ the quotient $2c\sigma t/(D + 2c\sigma t)$ increases with $t \le \rho$, so it is at most $r/(1+r)$ with

$$r = \frac{2c\sigma\rho}{D} = \frac{\rho \sin 2\theta}{\beta + 1 + (\beta - 1)\cos 2\theta} \le \frac{\rho}{\sqrt{(\beta+1)^2 - (\beta-1)^2}} = \frac{\rho}{2\sqrt{\beta}},$$

using $\gamma = 2 - \beta$ and $\max_x \sin x/(p + q\cos x) = (p^2 - q^2)^{-1/2}$ for $p > |q|$. Finally $r^2 \le \rho^2/(4\beta) = \frac{1}{4}\big(2 - \beta - (1-\delta^2)/\beta\big)$, which is maximal at $\beta = \sqrt{1-\delta^2} = \cos\psi$ with the value $\frac{1}{2}(1 - \cos\psi) = \sin^2(\psi/2)$. All steps are equalities for v with $v^*Wv = \cos\psi$, a phase of $v_\perp$ with $v^*Wv_\perp = \rho > 0$, and the optimal θ, and $r \mapsto r/(1+r)$ is increasing. □

Proposition 4.4. *Let $\mathrm{Tr}\, W = n$. For item 11, $V(W) = 1 - \mathrm{GM}/\mathrm{AM}$ of the eigenvalues of W, attained at $A = W^{-1}$. Over the densities with contrast δ the maximum is attained at a two-level spectrum:*

$$V_n(\delta) = 1 - \min_{1 \le m < n} \frac{n\,\ell^{(n-m)/n}}{m + (n-m)\ell}, \qquad \ell = \frac{1}{h} = \frac{1-\delta}{1+\delta}.$$

In particular $V_2 = 1 - \sqrt{1-\delta^2}$, and $V_n \uparrow 1 - 1/S(h)$ as $n \to \infty$, where $S(h) = \frac{(h-1)h^{1/(h-1)}}{e \log h}$ is the Specht ratio.

Proof. By the AM–GM inequality for the eigenvalues of $W^{1/2}AW^{1/2}$, $\varphi_W(A)/n \geq (\det W \det A)^{1/n}$, with equality for $A = W^{-1}$, so $\inf_{A>0} \varphi_W(A)/(n(\det A)^{1/n}) = (\det W)^{1/n}$ and $V(W) = 1 - (\det W)^{1/n}$, which for $\operatorname{Tr} W = n$ is $1-\mathrm{GM/AM}$. In the coordinates $x_i = \log w_i$, $\mathrm{AM/GM} = \frac{1}{n}\sum_i e^{x_i - \bar{x}}$ is convex, so on the box of spectra in $[b, a]$ it is maximal at a vertex, a two-level spectrum with m eigenvalues a and $n - m$ eigenvalues b; this gives the formula. Its limit is Specht's theorem [29]: $\sup \mathrm{AM/GM}$ over all finite families in $[b, a]$ is $S(h)$. □

Proposition 4.5. *Let* $\operatorname{Tr} W = n$. *For item 24,* $V(W) = w_{\min}^{-1/2} - 1$, *and*

$$V_n(\delta) = \sqrt{\frac{\ell + n - 1}{n\ell}} - 1, \qquad \ell = \frac{1-\delta}{1+\delta};$$

$V_2 = (1-\delta)^{-1/2} - 1 \sim \delta/2$, $V_n \sim \frac{n-1}{n}\delta$, *and* $V_n \uparrow \sqrt{h} - 1$ *as* $n \to \infty$.

Proof. For $A \geq 0$, $\varphi(A^2) = \operatorname{Tr}(A^{1/2}WA^{1/2}A) \leq \|A\|\varphi(A) \leq \operatorname{Tr}(A)\varphi(A) \leq \varphi(A)^2/w_{\min}$, with equality for the projection onto an eigenvector for $w_{\min}$. With $\operatorname{Tr} W = n$ and the extreme eigenvalues in the ratio h, $w_{\min} = nb/\sum_i w_i \geq nb/(b + (n-1)a) = n\ell/(\ell + n - 1)$, with equality for the spectrum $\{a, \ldots, a, b\}$. For $n = 2$, $(\ell + 1)/(2\ell) = 1/(1-\delta)$. □

Remark 4.6. Items 11 and 24 characterise the canonical trace among the functionals with $\varphi(1) = n$, so they are violated also by central densities on a direct sum (Section 7), and their envelopes grow with n, unlike all others with a closed form.

5 Inequalities for two projections

Theorem 5.1. *Let* $W > 0$ *in* M_n, $n \geq 2$, *with* $\Delta(W) = \delta$, *and let* P, Q *be projections. With* N *as in Table 1:*

(17) $V_n = \kappa^2 - 1$ *for* $\varphi(PQP) \leq \varphi(Q)$;

(18) $V_n = \kappa - 1$ *for* $\varphi(|P - Q|) \leq \varphi(P + Q)$;

(19) $V_n = \delta$ *for* $\varphi(P + Q) \leq \varphi(1 + P \wedge Q)$ *if* n *is even, and* $V_n = (n-1)\delta/(n-\delta)$ *if* n *is odd;*

(20) for $\varphi(e^{QPQ}) \leq \varphi(e^P)$ *and even* n, $V_n = \max_{0<g<\pi} \lambda(\delta, g)$, *where* $\lambda(\delta, g)$ *is the larger root of the quadratic* (2) *below, and* $V_n = \frac{(e-1)^2}{e(e+1)}\delta^2 + O(\delta^3)$;

(21) $V_n = \frac{1}{2}\tan^2(\psi/2)$ *for* $\varphi(\operatorname{Re} PQ) \geq 0$;

(22) $V_n = \kappa - 1$ *for* $\varphi(\operatorname{Re} PQ) \leq \varphi(|PQ|)$;

(23) $V_n = \delta/(1-\delta)$ *for* $\varphi(P \vee Q) \leq \varphi(P) + \varphi(Q)$.

The values are attained in 17, 18, 20–22 by rank-one P, Q, *and approached in 19 and 23 as* $P \neq Q$ *tend to the same eigenvector of* W. *For odd* n, V_n *in item 20 lies between the* M_2 *value and the same maximum with* $(1-\delta)/(n-1)$ *added to* β *in* (2).

Proof. Step 1: reduction to rank-one projections on M_2. By Lemma 2.2, $\mathbb{C}^n = \bigoplus_j H_j$ with $\dim H_j \in \{1, 2\}$, every H_j invariant under P and Q. Every operator in L, R and N (namely PQP, Q, $|P - Q|$, $P + Q$, $1 + P \wedge Q$, e^{QPQ}, e^P, $\operatorname{Re} PQ$, $|PQ|$, $P \vee Q$) is then block diagonal, with the block obtained from P_j, Q_j; for the lattice operations because $\operatorname{ran} P \cap \operatorname{ran} Q = \bigoplus_j (\operatorname{ran} P_j \cap \operatorname{ran} Q_j)$, the projections onto the H_j commuting with P and Q. Hence $L = \sum_j L_j$, $R = \sum_j R_j$ and $N = \sum_j N_j$, where L_j is

the same expression for P_j, Q_j and the functional $\mathrm{Tr}(W_j\,\cdot\,)$ on H_j, with the compression W_j of W to H_j; its spectrum lies in $[b,a]$, so $\Delta(W_j)\le\delta$. On a one-dimensional H_j, $P_j, Q_j \in \{0,1\}$ commute and $L_j \le R_j$ in all seven items (e.g. $\max(p,q)\le p+q$, $p+q\le 1+pq$, $e^{pq}\le e^p$). On a two-dimensional H_j, $P_j \ne Q_j$ are rank-one projections. Since $N_j \ge 0$, and $L_j \le R_j$ wherever $N_j = 0$,

$$\frac{L-R}{N} \le \max_j \frac{L_j - R_j}{N_j},$$

and the quotients do not change when W_j is scaled. It therefore suffices to bound $(L-R)/N$ for $W = \frac12(1+\tau\sigma_3)$, $0\le\tau\le\delta$, and $P = \frac12(1+\mu\cdot\sigma)$, $Q = \frac12(1+\nu\cdot\sigma)$ with unit Bloch vectors $\mu \ne \nu$ in $\mathbb{R}^3$ ($\sigma = (\sigma_1,\sigma_2,\sigma_3)$ the Pauli matrices). For fixed μ,ν the quotient is a linear-fractional function of τ (all three quantities are affine in W) with positive denominator, hence monotone on $[-\delta,\delta]$; conjugation by σ_1 maps τ to $-\tau$ and (μ,ν) to reflected vectors. So the bound for $\tau=\delta$ covers every $\tau\le\delta$, and the M_2 envelope bounds the envelope in every dimension. In 17, 18 and 21–23, N vanishes on blocks where $P = Q = 0$, so the values below, attained (or approached) on M_2, are attained (approached) on M_n with P, Q supported on the span of e_1, e_2. In 19 and 20, N contains $\varphi(1)$, which we treat at the end.

Step 2: formulas on M_2. Write $\varphi(x_0 + x\cdot\sigma) = x_0 + \delta x_3$, $\mu\cdot\nu = \cos 2\theta$ $(0<\theta\le\pi/2)$ and $c = \cos\theta = |\langle u,v\rangle|$ for unit vectors u, v with $P = uu^*$, $Q = vv^*$. Then $PQP = c^2P$, $(P-Q)^2 = \sin^2\theta\cdot 1$, $P\wedge Q = 0$, $P\vee Q = 1$, $\mathrm{Re}\,PQ = \frac14(1+\mu\cdot\nu+(\mu+\nu)\cdot\sigma)$, $|PQ| = (QPQ)^{1/2} = cQ$, $e^{QPQ} = 1+(e^{c^2}-1)Q$, $e^P = 1+(e-1)P$, and $|\mu+\nu| = 2\cos\theta$. We use the following elementary fact: *for $\beta > |z|$, $\max_{|\xi|=1}(\alpha+y\cdot\xi)/(\beta+z\cdot\xi)$ is the unique zero λ of $\alpha-\lambda\beta+|y-\lambda z|$*, since $(\alpha+y\cdot\xi)-\lambda(\beta+z\cdot\xi)\le\alpha-\lambda\beta+|y-\lambda z|$ with equality for $\xi\parallel y-\lambda z$, and the right-hand side decreases strictly in λ.

(17) $L/R = (1+\mu\cdot\nu)(1+\delta\mu_3)/(2(1+\delta\nu_3))$. For fixed μ, the fact with $\alpha=\beta=1$, $y=\mu$, $z=\delta e_3$ gives $\max_\nu(1+\mu\cdot\nu)/(1+\delta\nu_3) = 2(1-\delta\mu_3)/(1-\delta^2)$, so $L/R \le (1-\delta^2\mu_3^2)/(1-\delta^2) \le \kappa^2$, with equality for $\mu_3 = 0$ and $\nu = (\mu-\lambda\delta e_3)/(\lambda-1)$, $\lambda = 2/(1-\delta^2)$.

(18) $L/R = \sin\theta/(1+\frac{\delta}{2}(\mu_3+\nu_3)) \le \sin\theta/(1-\delta\cos\theta) \le \kappa$, the maximum being at $\cos\theta=\delta$, $\mu+\nu\parallel -e_3$. (Alternatively, $\varphi(|P-Q|)\le\kappa\varphi(P+Q)$ is the triangle inequality of Theorem 3.1 with $X=P$, $Y=-Q$.)

(19) $(L-R)/N = \varphi(P+Q)-1 = \frac{\delta}{2}(\mu_3+\nu_3) < \delta$, approached as $\mu,\nu\to e_3$.

(20) For fixed $g = 2\theta$, the pair (μ_3,ν_3) ranges over the image of the unit sphere under $\xi\mapsto(\mu\cdot\xi,\nu\cdot\xi)$, a filled ellipse, and the linear-fractional quotient is maximal on its boundary, $\mu_3 = \cos x$, $\nu_3 = \cos(x+g)$. With $K = e-1$ and $E = e^{\cos^2(g/2)}-1$,

$$\frac{L-R}{N} = \frac{\alpha+y\cdot\xi_x}{\beta+z\cdot\xi_x},\quad \alpha = \tfrac{E-K}{2},\ y = \tfrac{\delta}{2}(E\cos g - K,\ -E\sin g),\ \beta = 1+\tfrac{K}{2},\ z = (\tfrac{\delta K}{2},0),$$

$\xi_x = (\cos x,\sin x)$. By the fact, its maximum $\lambda(\delta,g)$ is the larger root of

$$(\beta^2-|z|^2)\lambda^2 - 2(\alpha\beta - y\cdot z)\lambda + \alpha^2 - |y|^2 = 0. \tag{2}$$

For small δ the maximiser has $g = 2K\delta/e + O(\delta^2)$, $x=\pi/2$, and $L-R = -\frac{e}{8}g^2 + \frac{K\delta}{2}g + O(\delta^3)$, which gives $V = K^2\delta^2/(e(e+1)) + O(\delta^3)$.

(21) With $s = \mu_3+\nu_3 \ge -2\cos\theta$, $(L-R)/N = -(2\cos^2\theta+\delta s)/(4(1+\frac{\delta}{2}s))$, which decreases in s; so $(L-R)/N \le (\delta C - C^2)/(2(1-\delta C))$, $C = \cos\theta$. The maximum over C is at $\delta C^2 - 2C + \delta = 0$, $C = \tan(\psi/2)$, with the value $C^2/2 = \frac12\tan^2(\psi/2)$.

(22) $\varphi(\mathrm{Re}\,PQ) = \mathrm{Re}\,\varphi(PQ) \le |\varphi(PQ)| \le \kappa\varphi(|PQ|)$ by Theorem 3.1, in every dimension. Equality: $v = \sqrt{b/(a+b)}\,e_1 + \sqrt{a/(a+b)}\,e_2$, $u = Wv/\|Wv\|$; then $PQ = \langle u,v\rangle uv^*$ has the polar factor of the equality case of Theorem 3.1.

(23) $L/R = 1/(1+\frac{\delta}{2}(\mu_3+\nu_3)) < 1/(1-\delta)$, approached as $\mu,\nu \to -e_3$.

Step 3: items 19 and 20 in dimension n. For even n take W with the eigenvalues a and b, each $n/2$ times, and $n/2$ copies of the M_2 extremiser, one in each block spanned by an eigenvector for a and one for b; this attains (approaches) the M_2 value, which is an upper bound by Step 1. For odd n at least one block of Lemma 2.2 is one-dimensional, so there are $k' \le k = (n-1)/2$ two-dimensional blocks. In item 19, $N \ge \varphi(1) = \operatorname{Tr} W$; by Step 2 a two-dimensional block contributes $L_j - R_j = \operatorname{Tr} W_j \cdot \frac{\Delta(W_j)}{2}(\mu_3+\nu_3) < \lambda_1(W_j) - \lambda_2(W_j)$, and a one-dimensional one $L_j - R_j \le 0$. By Ky Fan's maximum principle [3, Sect. III.1], $\sum_j \lambda_1(W_j)$ is at most the sum of the k' largest and $\sum_j \lambda_2(W_j)$ at least the sum of the k' smallest eigenvalues of W. The resulting bound for $(L-R)/N$ is a linear-fractional function of the spectrum of W and is maximal for $k' = k$ and the spectrum $a^{(k)}, b^{(k+1)}$; hence $(L-R)/N < k(a-b)/(k(a+b)+b) = (n-1)\delta/(n-\delta)$, which is approached for this spectrum with $P = 1$, $Q = 0$ on the remaining eigenvector (a block with $L_j = R_j$). In item 20 the same configuration, with $P = Q = 0$ on the remaining eigenvector, adds b/k per block to N, i.e. $(1-\delta)/(n-1)$ to β (normalisation $\operatorname{Tr} W_j = 1$); this gives the lower bound, and the M_2 value the upper bound. □

6 Lower bounds

Proposition 6.1. *For every $n \ge 2$:*

(a) $V_n \ge \kappa - 1$ for item 6 (approached);

(b) $V_n \ge \kappa^2 - 1 = \delta^2/(1-\delta^2)$ for item 7 (approached);

(c) $V_n \ge m(\delta) := \max_{0<y<\delta} \dfrac{y^2(\delta-y)^2}{(1-2\delta y+y^2)(1-y^2)} = \dfrac{\delta^4}{16} + O(\delta^5)$ for item 10 (attained);

(d) $V_n \ge \delta^2/36 + O(\delta^3)$ for item 14, and more generally $(1-\frac{2}{p})^2\delta^2/4 + O(\delta^3)$ for Hölder's inequality $|\varphi(PQ)| \le \varphi(P)^{1/p}\varphi(Q)^{1/q}$.

By Lemma 2.3, $V_n \le h - 1$ in (a)–(c).

Proof. All constructions live on $\operatorname{span}\{e_1, e_2\}$ (extended by 0, or by a small multiple of the identity where positive definiteness is required); let $e_\pm = (e_1 \pm e_2)/\sqrt{2}$.

(a) For invertible A and a rank-one $B = vv^*$, $\|v\| = 1$, we have $A^{-1}\#vv^* = (v^*Av)^{-1/2}vv^*$, hence $(A^{-1}\#B)^{1/2} = (v^*Av)^{-1/4}vv^*$ and $A\natural B = (v^*Av)^{1/2}vv^*$. With $v = e_+$ and $A = xx^* + \varepsilon$,

$$\frac{\varphi(A\natural B)^2}{\varphi(A)\varphi(B)} = \frac{(v^*Av)(v^*Wv)}{\operatorname{Tr}(WA)} \xrightarrow[\varepsilon\to 0]{} \frac{|v^*x|^2\, v^*Wv}{x^*Wx},$$

which for $x = W^{-1}v$ equals $(v^*W^{-1}v)(v^*Wv) = \frac{1}{4}(a^{-1}+b^{-1})(a+b) = \kappa^2$. By continuity the quotient L/R approaches κ for positive definite A, $B = vv^* + \varepsilon'$.

(b) In the basis $f_1 = e_+$, $f_2 = e_-$ the density has the entries $W_{11} = W_{22} = \frac{a+b}{2}$, $W_{12} = \frac{a-b}{2}$. Take $B = \eta f_1 f_1^* + f_2 f_2^*$ and $A = uu^*$ with $u = f_1 + s\eta f_2$. Then $B^{1/2}AB^{1/2} = ww^*$ with $w = \sqrt{\eta} f_1 + s\eta f_2$, so $L = \|w\|^2\, w^*Ww = \eta^2 W_{11} + O(\eta^{5/2})$, and $Bu = \eta(f_1 + sf_2)$, so $R = \|u\|^2\eta^2(W_{11} + 2sW_{12} + s^2W_{22})$. As $\eta \to 0$, $L/R \to W_{11}/(W_{11} + 2sW_{12} + s^2W_{22})$, which at $s = -W_{12}/W_{22}$ equals $W_{11}W_{22}/(W_{11}W_{22} - W_{12}^2) = (a+b)^2/(4ab) = \kappa^2$.

(c) Normalise $a + b = 2$, so $W = \operatorname{diag}(1+\delta, 1-\delta)$, and take $A = uu^*$, $B = A + tvv^*$, $t \ge 0$, with $u = (\cos\alpha, \sin\alpha)$, $v = (-\cos\alpha, \sin\alpha)$ and $y = -\cos 2\alpha$. Then $u^*v = y$, $v^*Wu = y - \delta$ and $u^*Wu = v^*Wv = 1 - \delta y$, so

$$L - R = \varphi(A^2) - \varphi(B^2) = -2t\operatorname{Re}\big((u^*v)(v^*Wu)\big) - t^2 v^*Wv = 2ty(\delta - y) - t^2(1-\delta y).$$

For $0 < y < \delta$ the best $t = y(\delta - y)/(1 - \delta y) > 0$ gives $L - R = G := y^2(\delta - y)^2/(1 - \delta y)$ and $(L - R)/R = G/(1 - \delta y - G) = y^2(\delta - y)^2/((1 - 2\delta y + y^2)(1 - y^2))$. Maximising over y gives $m(\delta)$; the leading order comes from $y = \delta/2$.

(d) Take $W = \frac{1}{2}(1 + \delta D)$, $D = \mathrm{diag}(1, -1)$, and real unit vectors u, v at the angles $m \pm \varepsilon/2$ to e_1, $P = uu^*$, $Q = vv^*$. Then $\varphi(PQ) = \frac{1}{2}(\cos^2\varepsilon + \delta\cos\varepsilon\cos 2m)$ and $\varphi(P)^{1/p}\varphi(Q)^{1/q} = \frac{1}{2}\big(1 + \delta\cos 2m + \delta(\frac{1}{q} - \frac{1}{p})\varepsilon\sin 2m\big) + O(\delta^2\varepsilon^2)$, so $L - R = \frac{1}{2}\big(-\varepsilon^2 - \delta(1 - \frac{2}{p})\varepsilon\sin 2m\big) + O(\delta^3)$ for $\varepsilon = O(\delta)$. The choice $m = \pi/4$, $\varepsilon = -\frac{\delta}{2}(1 - \frac{2}{p})$ gives $L - R = \frac{1}{8}(1 - \frac{2}{p})^2\delta^2 + O(\delta^3)$ and $R = \frac{1}{2} + O(\delta)$. □

Proposition 6.2. *For item 25 ($\varphi(A^4) \le \varphi(AB^2A)$ for $0 \le A \le B$), rank-one A gives no violation, and an explicit construction with A of rank two and $B - A$ of rank one gives*

$$V_n \ge m_{25}(\delta) = \frac{\delta^8}{4096}\big(1 + O(\delta^2)\big) \qquad \text{in every dimension } n \ge 2.$$

Proof. If $A = \alpha uu^*$ has rank one and $A \le B$, then $u^*Bu \ge \alpha$ and $AB^2A = \alpha^2\|Bu\|^2uu^* \ge \alpha^2(u^*Bu)^2uu^* \ge A^4$, so every positive φ satisfies the inequality. For the construction take W with $\mathrm{Tr}\,W = 1$ and eigenvalues $(1 \pm \delta)/2$, and, in an orthonormal basis at the angle ϑ to its eigenbasis, $A = \mathrm{diag}(1, s)$, $0 < s < 1$, and $B = A + tvv^*$ with $v = (\cos\phi, \sin\phi)$; in this basis W has the entries $w_{11,22} = \frac{1}{2}(1 \pm \delta\cos 2\vartheta)$ and $w_{12} = \frac{\delta}{2}\sin 2\vartheta$. Then $R - L = tT_1 + t^2T_2$ with $T_1 = \varphi(A^2P_vA + AP_vA^2)$, $T_2 = \varphi(AP_vA)$, $P_v = vv^*$, and whenever $T_1 < 0$ the best $t = -T_1/(2T_2)$ gives $(L - R)/R = G/(L - G)$, $G = T_1^2/(4T_2)$. Let $m_{25}(\delta)$ be the maximum over (ϑ, s, ϕ); it is attained, and by embedding $\mathbb{C}^2 \subset \mathbb{C}^n$ (with A, B extended by 0) it is a lower bound in every dimension. For the leading order put $\vartheta = \pi/4$, $\sqrt{s} = \sigma\delta$ and $\tan\phi = -\varrho s^{-3/2}$. Then $T_1 = \cos^2\phi\,(1 + \varrho^2 - (1 + s)\varrho/\sigma)$, $T_2 = \cos^2\phi\,(\frac{1}{2} + \frac{\varrho^2}{2s} - \frac{\varrho}{\sigma})$ and $\cos^2\phi = s^3/(s^3 + \varrho^2)$, so that

$$\frac{L - R}{R} = \big(\sigma^2(\varsigma - \varsigma^2) - \sigma^4\big)^2\delta^8 + O(\delta^{10}), \qquad \varsigma = \sigma/\varrho,$$

which is maximal for $\varsigma = \frac{1}{2}$, $\sigma^2 = \frac{1}{8}$, with the value $\delta^8/4096$. □

The numerical maximisation gives $m_{25}(0.1)/(0.1^8/4096) = 1.020$. The same scaling for a general exponent $r \ge 0$ in $\varphi(A^{2r+2}) \le \varphi(A^rB^2A^r)$ gives the order $\delta^{4(r+1)}$; for $r = 0$ the supremum of the leading coefficient is $\frac{1}{16}$, that of item 10.

Remark 6.3. With $N = R$, item 13 would have $V = \infty$ for every $\delta > 0$: for rank-one $A = uu^*$, $B = vv^*$ with overlap $c = |u^*v| \to 0$, $L \approx 2c\,|u^*Wu_\perp|$ while $R = 2c^2u^*Wu$. This is why we normalise by $\varphi(A + B)$.

7 Direct sums: only the contrast inside the blocks matters

Let $\mathcal{M} = \bigoplus_{c=1}^{C} M_{k_c}$, with the trace $\mathrm{Tr} = \sum_c \mathrm{Tr}_c$ and $\varphi_W = \mathrm{Tr}(W\,\cdot\,)$, $W = \bigoplus_c W_c \ge 0$. The traces on $\mathcal{M}$ are the functionals $\sum_c t_c\,\mathrm{Tr}_c$, $t_c \ge 0$; so φ_W is a trace iff W is central, whatever its global contrast. The global contrast $\Delta(W)$ thus says nothing about traciality, and the following theorem shows that it says nothing about the size of the violation either.

Theorem 7.1. *Let $W = \bigoplus_c W_c > 0$ in $\mathcal{M}$, and let $V_{\mathcal{M}}(W)$ and $V_{M_{k_c}}(W_c)$ be the suprema of the normalised violation of one inequality of Table 1, other than 11, 16 and 24, for φ_W on $\mathcal{M}$ and for φ_{W_c} on M_{k_c}. Then*

$$V_{\mathcal{M}}(W) \le \max_c V_{M_{k_c}}(W_c),$$

with equality for items 1–10, 12–14, 17, 18, 21–23 and 25. In particular, if every W_c is scalar, no inequality of Table 1 except 11 and 24 is violated, however different the scalars are; and the envelope

on $\mathcal{M}$ along the intra-block contrast $\max_c \Delta(W_c) = \delta$ is at most the largest envelope $V_{k_c}(\delta)$ of the blocks.

Proof. All variables of the items are in $\mathcal{M}$, hence block diagonal, and every functional calculus acts blockwise. Let L_c, R_c, N_c be the expressions of the item for the block variables and φ_{W_c}. For all items except 11, 16 and 24, $L \le \sum_c L_c$ (with equality unless L contains an absolute value of a functional, as in 1, 4 and 14), $N = \sum_c N_c$, and $R = \sum_c R_c$, except in items 6 and 14, where $R = \sqrt{\sum_c \varphi_c(A_c) \sum_c \varphi_c(B_c)} \ge \sum_c R_c$ by the Cauchy–Schwarz inequality and $R \ge \sum_c R_c$ by Hölder's inequality. Since $W > 0$, $N_c = 0$ forces the variables of block c to vanish where they enter L_c, so $L_c \le R_c$ there. Hence $(L - R)/N \le \sum_c (L_c - R_c) / \sum_c N_c \le \max_c (L_c - R_c)/N_c$, the maximum over the blocks with $N_c > 0$ (for $N = R$ in items 6 and 14 use $L/R \le \sum_c L_c / \sum_c R_c$ instead). This proves the inequality. For the equality, in the listed items N is additive and the variables may vanish on all blocks but one without making N vanish; restricting a near-optimal configuration of one block and setting the variables to zero on the other blocks reproduces its quotient (in 19, 20 and 15, N contains $\varphi(1)$ and the other blocks dilute the violation). If every W_c is scalar, every $V_{M_{k_c}}(W_c)$ vanishes, since φ_{W_c} is then a trace on M_{k_c}; for item 16 the same conclusion holds because the Peierls–Bogoliubov inequality holds for every trace on $\mathcal{M}$. □

Remark 7.2. The theorem is a statement about the characterisations, not about the inequalities: a non-central density is detected by every characterisation of traces, but only through the variation *inside* the blocks. Items 11 and 24 characterise the *canonical* trace Tr among the functionals with $\varphi(1) = \dim$, and central W violate them; for a fixed density their violations are $1 - \mathrm{GM/AM}$ and $w_{\min}^{-1/2} - 1$ of the eigenvalues of W normalised to $\operatorname{Tr} W = \sum_c k_c$ (Propositions 4.4 and 4.5; the proofs do not use the block structure).

8 Gardner's inequality with two prescribed contrasts

In [1, Sect. 5] the violation of Gardner's inequality was studied without normalisation: for $X \ge 0$ with $\|X\| = 1$ (the density) and Z with $\|Z\| = 1$, the quantity

$$z(X, Z) = |\operatorname{Tr}(XZ)| - \operatorname{Tr}(X|Z|)$$

was sampled against $\Delta(X)$ and $\Delta(|Z|)$. Writing $Z = U|Z|$ with a unitary (orthogonal, in the real case) U, $z = |\operatorname{Tr}(XUY)| - \operatorname{Tr}(XY)$ with $Y = |Z| \ge 0$, and $\sup_U z = \|YX\|_1 - \operatorname{Tr}(XY)$ by the duality of the trace norm, the supremum being attained at the polar unitary of YX, which is real if X, Y are. We first solve the 2×2 case.

Proposition 8.1. *Let $X, Y \ge 0$ in $M_2(\mathbb{C})$ have eigenvalues $x_1 \ge x_2 \ge 0$ and $y_1 \ge y_2 \ge 0$, $x_1, y_1 > 0$. For every unitary U,*

$$|\operatorname{Tr}(XUY)| - \operatorname{Tr}(XY) \;\le\; \frac{(x_1 - x_2)^2 (y_1 - y_2)^2}{4(x_1 + x_2)(y_1 + y_2)}.$$

Let $S = x_1 y_2 + x_2 y_1$, $q = (x_1 + x_2)(y_1 + y_2)$, and $\gamma = |\langle e_1, f_1 \rangle|^2$ for unit eigenvectors e_1 of X for x_1 and f_1 of Y for y_1. If $x_1 > x_2$ and $y_1 > y_2$, the bound is attained iff $|\operatorname{Tr}(XUY)| = \|YX\|_1$ and $\gamma = \frac{1}{4} + \frac{S}{2q}$; it is attained by real X, Y, U.

Proof. Let $p = (x_1 - x_2)(y_1 - y_2)$, so that $q - 2S = p$. First, $|\operatorname{Tr}(XUY)| = |\operatorname{Tr}(YXU)| \le \|YX\|_1$. Expanding in the eigenbases, $\operatorname{Tr}(XY) = \sum_{ij} x_i y_j |\langle e_i, f_j \rangle|^2 = S + \gamma p$, since the matrix $(|\langle e_i, f_j \rangle|^2)$ is doubly stochastic. The singular values of YX satisfy $s_1^2 + s_2^2 = \operatorname{Tr}(X^2 Y^2)$ and $s_1 s_2 = |\det YX| =$

$x_1x_2y_1y_2$, so $\|YX\|_1^2 = \operatorname{Tr}(X^2Y^2) + 2x_1x_2y_1y_2 = S^2 + \gamma pq$. With $\sqrt{r} \le r/q + q/4$ (equality iff $\sqrt{r} = q/2$),

$$\|YX\|_1 - \operatorname{Tr}(XY) \le \frac{S^2 + \gamma pq}{q} + \frac{q}{4} - S - \gamma p = \frac{(q/2 - S)^2}{q} = \frac{p^2}{4q},$$

with equality iff $\gamma pq = q^2/4 - S^2 = \frac{p}{2}(\frac{q}{2} + S)$. Every $\gamma \in [0,1]$ is realised by a real rotation, and the polar unitary of a real YX is real. □

Theorem 8.2. *Let $\mathcal{M} = \bigoplus_{c=1}^{C} M_{k_c}(\mathbb{K})$, $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, and $N = \sum_c \lfloor k_c/2 \rfloor \ge 1$. For $X \ge 0$ and $Y \ge 0$ in $\mathcal{M}$ with $\|X\| = \|Y\| = 1$, $\Delta(X) = \delta \in [0,1)$ and $\Delta(Y) = \delta' \in [0,1]$, and every unitary (orthogonal) $U \in \mathcal{M}$,*

$$|\operatorname{Tr}(XUY)| - \operatorname{Tr}(XY) \le \frac{N\,\delta^2\delta'^2}{(1+\delta)(1+\delta')},$$

and the bound is attained. Equivalently, $\sup\{|\operatorname{Tr}(XZ)| - \operatorname{Tr}(X|Z|) : \|X\| = \|Z\| = 1,\ \Delta(X) = \delta,\ \Delta(|Z|) = \delta'\}$ *has this value.*

Proof. Let $\ell = (1-\delta)/(1+\delta)$ and $\ell' = (1-\delta')/(1+\delta')$ be the smallest eigenvalues. It suffices to prove the bound in the complex algebra, since the maximisers below are real. By the remark above, $\sup_U z = F(X,Y) := \|YX\|_1 - \operatorname{Tr}(XY)$, and the polar unitary lies in $\mathcal{M}$. For a fixed eigenbasis of X, F is a convex function of the eigenvalues of X (a norm of a linear function of them minus a linear function). The admissible eigenvalues form the union of the faces $\{x_i = 1,\ x_j = \ell\}$ of the cube $[\ell,1]^K$, $K = \sum_c k_c$, and a convex function on a face attains its maximum at a vertex; so $F(X,Y) \le F(X',Y)$ for some $X' = \ell + (1-\ell)P$ with a projection $P \in \mathcal{M}$, $P \ne 0,1$, and likewise $F(X',Y) \le F(X',Y')$ with $Y' = \ell' + (1-\ell')Q$. By Lemma 2.2, applied in every block, X' and Y' are direct sums over subspaces of dimension 1 or 2, and so are $\|Y'X'\|_1$ and $\operatorname{Tr}(X'Y')$. One-dimensional summands contribute 0. On a two-dimensional one, $X' = \operatorname{diag}(1,\ell)$ and Y' has the eigenvalues $1, \ell'$, so by Proposition 8.1 it contributes at most $(1-\ell)^2(1-\ell')^2/(4(1+\ell)(1+\ell')) = \delta^2\delta'^2/((1+\delta)(1+\delta'))$. A block of size k_c holds at most $\lfloor k_c/2 \rfloor$ two-dimensional summands. Equality: N blocks $\operatorname{diag}(1,\ell)$ and Y with spectrum $\{1,\ell'\}$ at the angle of Proposition 8.1, $X = Y = 1$ on the remaining one-dimensional blocks, and the polar unitary, which makes all block traces positive. □

Remark 8.3. (i) With the contrast of $|Z|$ free, the supremum $N\delta^2/(2(1+\delta))$ is reached at $\delta' = 1$, i.e. for singular Z. (ii) The bound is additive over the blocks: it counts the mutually orthogonal two-dimensional subspaces that fit into the blocks, and it does not depend on the sizes of the blocks otherwise. (iii) The normalised version is Theorem 3.1: $\sup_Z |\operatorname{Tr}(XZ)|/\operatorname{Tr}(X|Z|) = \kappa(X)$, attained by a rank-one $|Z|$, i.e. at $\Delta(|Z|) = 1$. The unnormalised supremum vanishes to fourth order at $\delta = \delta' = 0$ because the violation needs both operators to be far from scalar. (iv) Figures 1 and 2 compare the bound with random sampling and with numerical optimisation. Sampling, the method of [1], stays far below the bound as soon as the blocks are larger than 2×2, while the optimisation over both contrasts reproduces it.

9 Numerical verification and the remaining items

9.1 Method

All envelopes were computed by constrained optimisation with our PyTorch library `torch_vn_algebra` [24]; the program is `scripts/trace_characterizations.py` of the repository [24]. The density W is diagonal with a spectrum parametrised so that $\Delta(W) = \delta$ holds exactly and the other eigenvalues are free. Positive variables have a free contrast in $[0,1]$, so rank-deficient operators are

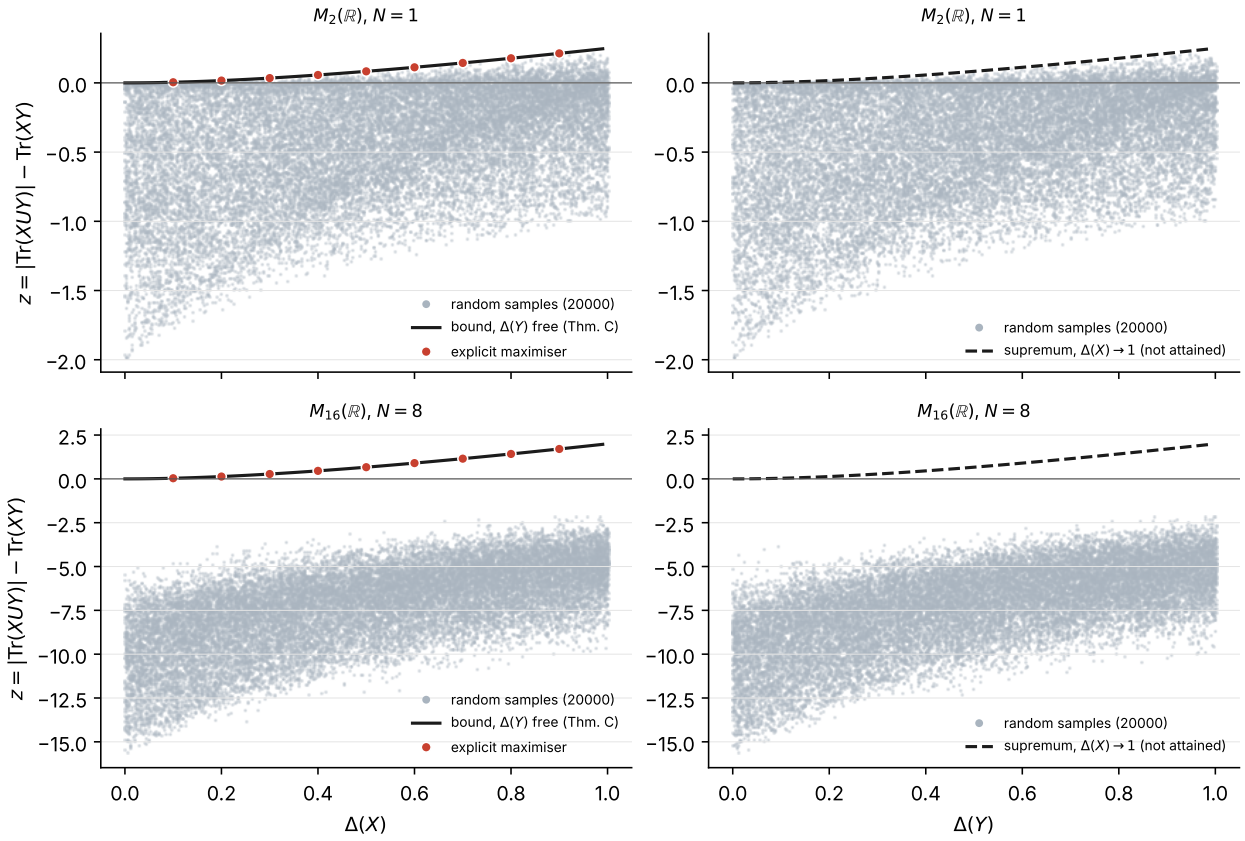


Figure 1: The unnormalised Gardner violation $z = |\operatorname{Tr}(XUY)| - \operatorname{Tr}(XY)$ for $\|X\| = \|Y\| = 1$ in $M_2(\mathbb{R})$ (top, $N = 1$) and $M_{16}(\mathbb{R})$ (bottom, $N = 8$), against $\Delta(X)$ (left) and $\Delta(Y)$ (right). Grey: 20000 random samples with Haar-orthogonal eigenbases and U and uniformly drawn contrasts, as in [1, Fig. 1]. Solid: the bound of Theorem 8.2 with $\Delta(Y)$ free, $N\delta^2/(2(1+\delta))$; dots: z at the explicit maximisers of its proof, which agree with it to 10^{-15}. Dashed: the supremum along $\Delta(Y)$, approached as $\Delta(X) \to 1$. In M_{16} no sample violates Gardner's inequality ($z > 0$), although violations up to almost 2 exist: the maximisers, two-level spectra on aligned pairs of eigenvectors, form a set of negligible Haar measure. Computed with plain NumPy by `scripts/trace_envelopes_figures.py`.

included; there are self-adjoint and unitary variables (polar factors, and projections UDU^*, with every rank pattern spread over the starts), and positive scalars. Half of the starts begin at structured points (two-level spectra, rotations in pairs of eigenvectors of W). The lattice operations are the spectral projections of $P + Q$ for the eigenvalue 2 ($P \wedge Q$) and for the range ($P \vee Q$), computed by functional calculus with an indicator function, whose Daleckii–Krein gradient is the derivative of the spectral projection. In 19 and 23 the suprema are approached as P and Q merge, where $P \wedge Q$ and $P \vee Q$ jump; there the search maximises $(L - R)/N - 10^{-12}/\|P - Q\|_{\mathrm{F}}^2$, which keeps P and Q about 10^{-3} apart, while the reported values are those of $(L - R)/N$. A grid of 14 contrasts times 64 starts forms one batch per item and dimension (800 gradient steps, $n = 2, 3, 4$).

Certification. The best final point of every contrast is recomputed in mpmath with 40 digits, after the unitaries are re-orthonormalised and the contrast is recomputed from the stored eigenvalues; the lattice operations are recomputed from the exact eigenvalues. Every positive value reported below is thus a certified violation at an exactly admissible point. The double-precision and mpmath values agree to 5×10^{-6} (relative), and at $\delta = 0$ no value exceeds 1×10^{-16}. Independently, plain

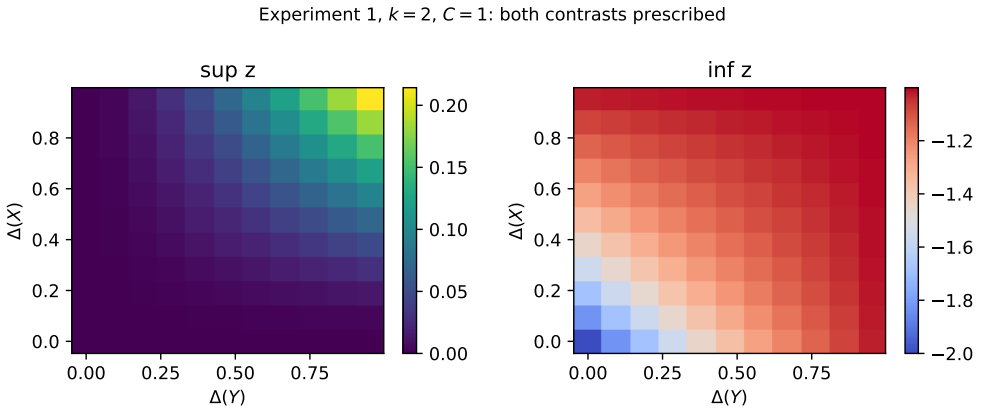


Figure 2: $M_2(\mathbb{R})$, both contrasts prescribed: the supremum (left) and infimum (right) of $z = |\operatorname{Tr}(XUY)| - \operatorname{Tr}(XY)$ found by constrained optimisation on a grid of $\Delta(Y)$ (horizontal) and $\Delta(X)$ (vertical). The left panel agrees with $\delta^2\delta'^2/((1+\delta)(1+\delta'))$ (Theorem 8.2) to 5×10^{-10} at every grid point: a violation needs both operators to be far from scalar.

NumPy/SciPy searches (BFGS from 40 random starts on M_2 at $\delta = 0.3, 0.5, 0.9$) give values whose ratio to the batched search lies in $[0.994, 1.097]$; ratios above 1 come from item 13, where BFGS stops earlier. The test suite compares every objective with a plain NumPy implementation and with mpmath, checks that random points stay below the proved envelopes and that the stated extremisers attain them. The whole computation takes 475 s on 4 CPU processes.

9.2 Results

items	$V^\star(\delta)$	as $\delta \to 0$	$V^\star(0.5)$	search $V(0.5)$	status
8	$h - 1 = 2\delta/(1-\delta)$	2δ	2.0000	2.0000	sharp
9	$s/(1+s)$, $s = \sin\frac{\psi}{2}$	$\delta/2$	0.2056	0.2056	sharp for $n = 2$
19	δ	δ	0.5000	0.5000	sharp, approached
23	$\delta/(1-\delta)$	δ	1.0000	1.0000	sharp, approached
24	$(1-\delta)^{-1/2} - 1$	$\delta/2$	0.4142	0.4142	sharp for $n = 2$; grows with n
1–5, 18, 22	$\kappa - 1$	$\delta^2/2$	0.1547	0.1547	sharp
17	$\kappa^2 - 1$	δ^2	0.3333	0.3333	sharp
11	$1 - \sqrt{1-\delta^2}$	$\delta^2/2$	0.1340	0.1340	sharp for $n = 2$; grows with n
20	max. of one variable	$\frac{(e-1)^2}{e(e+1)}\delta^2$	0.0802	0.0802	sharp
21	$\frac{1}{2}\tan^2(\psi/2)$	$\delta^2/8$	0.0359	0.0359	sharp
6	$\geq \kappa - 1$	$\geq \delta^2/2$	0.1547	0.1430	lower bound
7	$\geq \kappa^2 - 1$	$\geq \delta^2$	0.3333	0.2653	lower bound
12–16	–	$c\,\delta^2$, $c \in [\frac{1}{36}, \frac{1}{2}]$	–	0.0086–0.1338	search
10	$\geq m(\delta)$	$\geq \delta^4/16$	0.0052	0.0052	lower bound
25	$\geq m_{25}(\delta)$	$\geq \delta^8/4096$	1.7×10^{-6}	1.5×10^{-6}	lower bound

Table 2: Classes of behaviour, ordered by the order of vanishing of the envelope at the trace. $V^\star$: the sharp value or, where marked $\geq$, a proved lower bound; "search": the certified search value on M_2 (a range over the items of a row). Items 11 and 24 are given for $n = 2$.

Tables 2 and 3 and Figure 3 summarise the computation.

(i) *Agreement with the theorems.* Where the envelope is known exactly and attained (items 1–5, 8, 11, and 9 for $n = 2$), the search reproduces it to 1×10^{-7} for $\delta \geq 0.05$ and $n = 2, 3, 4$; for items

	inequality $L \le R$	ref.	$V(0.1)$	$V(0.5)$	$V(0.9)$	$V^\star(0.5)$	order	slope	n-dep.
1	$\lvert\varphi(X)\rvert \le \varphi(\lvert X\rvert)$	[18]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	4×10^{-10}
2	$\varphi(\lvert X+Y\rvert) \le \varphi(\lvert X\rvert)+\varphi(\lvert Y\rvert)$	[18, 30]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	1×10^{-9}
3	$\varphi(A+A^*) \le 2\varphi(\lvert A^*\rvert)$	[10]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	5×10^{-10}
4	$\lvert\varphi(A^2)\rvert \le \varphi(A^*A)$	[27, 14]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	1×10^{-8}
5	$\varphi(\lvert P-R\rvert) \le \varphi(\lvert P-Q\rvert)+\varphi(\lvert Q-R\rvert)$	[11]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	1×10^{-8}
6	$\varphi(A\natural B) \le \sqrt{\varphi(A)\varphi(B)}$	[15]	n.r.	0.1430	1.1979	0.1547^b	δ^2	–	$n\ge3$
7	$\varphi((B^{1/2}AB^{1/2})^2) \le \varphi(BA^2B)$	[9]	0.0059	0.2653	3.4322	0.3333^b	δ^2	2.31	9×10^{-2}
8	$\varphi(\lvert AB^*\rvert) \le \frac12\varphi(\lvert A\rvert^2)+\frac12\varphi(\lvert B\rvert^2)$	[4]	0.2222	2.0000	18.00	2.0000	δ^1	1.17	9×10^{-12}
9	$\varphi(PQ+QP) \le 2\varphi(QPQ)$	[10]	0.0477	0.2056	0.3469	0.2056^a	δ^1	0.94	3×10^{-10}
10	$0\le A\le B \Rightarrow \varphi(A^2)\le\varphi(B^2)$	[5, 28, 22]	6.3×10^{-6}	0.0052	0.1825	0.0052^b	δ^4	4.04	3×10^{-8}
11	$(\det A)^{1/n} \le \frac1n\varphi(A)$, $\varphi(1)=n$	[12]	0.0050	0.1340	0.5641	0.1340	δ^2	2.01	yes
12	$\varphi((A+B)^{1/2}) \le \varphi(A^{1/2})+\varphi(B^{1/2})$	[31]	0.0050	0.1338	0.9678	–	–	2.00	5×10^{-4}
13	$\varphi(A+B-\lvert A-B\rvert) \le 2\varphi(A^{1/4}B^{1/2}A^{1/4})$	[23]	0.0048	0.0911	0.2197	–	–	1.93	4×10^{-1}
14	$\lvert\varphi(PQ)\rvert \le \varphi(P)^{1/3}\varphi(Q)^{2/3}$	[8, 10]	2.8×10^{-4}	0.0086	0.0711	–	–	2.03	4×10^{-3}
15	$\varphi(e^{P+Q}) \le \varphi(e^{P/2}e^Qe^{P/2})$	[8]	6.4×10^{-4}	0.0142	0.0420	–	–	1.98	2×10^{-1}
16	$\varphi(e^A)e^{\varphi(e^{A/2}Be^{A/2})/\varphi(e^A)} \le \varphi(e^{A+B})$	[9]	9.7×10^{-4}	0.0561	0.8890	–	–	2.35	2×10^{-3}
17	$\varphi(PQP) \le \varphi(Q)$	[7]	0.0101	0.3333	4.2632	0.3333	δ^2	2.04	7×10^{-9}
18	$\varphi(\lvert P-Q\rvert) \le \varphi(P+Q)$	[7]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	1×10^{-8}
19	$\varphi(P+Q) \le \varphi(1+P\wedge Q)$	[7]	0.1000	0.5000	0.9000	0.5000^c	δ^1	1.00	odd n
20	$\varphi(e^{QPQ}) \le \varphi(e^P)$	[8]	0.0029	0.0802	0.3115	0.0802	δ^2	2.02	odd n
21	$\varphi(\mathrm{Re}\,PQ) \ge 0$	[8]	0.0013	0.0359	0.1964	0.0359	δ^2	2.02	8×10^{-9}
22	$\varphi(\mathrm{Re}\,PQ) \le \varphi(\lvert PQ\rvert)$	[8]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	5×10^{-9}
23	$\varphi(P\vee Q) \le \varphi(P)+\varphi(Q)$	[27]	0.1111	1.0000	9.0000	1.0000^c	δ^1	1.17	4×10^{-7}
24	$\varphi(A^2)^{1/2} \le \varphi(A)$, $\varphi(1)=n$	[12]	0.0541	0.4142	2.1623	0.4142	δ^1	1.13	yes
25	$0\le A\le B \Rightarrow \varphi(A^4)\le\varphi(AB^2A)$	[22]	n.r.	1.5×10^{-6}	0.0017	1.7×10^{-6b}	δ^8	–	3×10^{-6}

Table 3: Certified search values on M_2 at $\delta = 0.1, 0.5, 0.9$ and the proved values $V^\star$; ^{a}: sharp for $n=2$ only, ^{b}: a lower bound, ^{c}: approached but not attained. "order": leading order of $V^\star$ as $\delta\to0$; "slope": $\log_2(V(0.2)/V(0.1))$ of the search; "n-dep.": the largest $|V_n/V_2-1|$ for $n=3,4$ and $\delta\ge0.1$ (for items 7 and 13 it reflects search shortfalls; items 19 and 20 differ for odd n, Theorem 5.1). "n.r.": no violation resolved by the search (*full* run on *CPU(4threads)*).

17, 18, 20–22 and 24 to 3×10^{-4}, and for the approached suprema of 19 and 23 to 2×10^{-6}. Apart from item 6 with $n\ge3$, no value exceeds the lower bounds of Section 6 (largest relative excess 1×10^{-16}). The values of 17, 18 and 21–23 on M_3 and M_4 agree with the M_2 values to 4×10^{-7}, as Theorem 5.1 predicts, and the search on M_3 finds the odd-n value 0.0694 of item 20 at $\delta=0.5$ (M_2: 0.0802).

(ii) *Items without a closed form.* All five items 12–16 fail quadratically, with $V\approx0.5\,\delta^2$ (12, 13), $\delta^2/36$ (14, equal to the bound of Proposition 6.1(d) to leading order), $0.064\,\delta^2$ (15) and $0.09\,\delta^2$ (16). Item 15 decreases with n, because the identity part of $e^{P/2}e^Qe^{P/2}$ enters N; the others give the same values for $n=2,3,4$. In item 16 the best points have large $\|A\|$, $\|B\|$, and the supremum is approached as the scale grows.

(iii) *Dimension.* For items 9 and 10 the search gives the same values for $n=3,4$ as for $n=2$, to 4×10^{-3}. Item 6 depends on n: for $n=2$ the search never exceeds $\kappa-1$, but for $n\ge3$ it certifies larger violations, $V\ge0.847$ at $n=4$, $\delta=0.8$, where $\kappa-1=0.667$; a stored point gives $V>2.1$ at $n=3$, $\delta=0.9$, where $\kappa-1=1.29$. We only know $\kappa-1\le V_6\le h-1$.

(iv) *Extremisers.* Except for items 6 ($n\ge3$), 11 ($A=W^{-1}$) and 25 (A of rank two with eigenvalue ratio $\approx\delta^2/8$), the extremisers are rank-one operators or matrix units in the span of eigenvectors for a and b, and they do not commute with W.

(v) *Limits of the search.* The suprema of 6 and 7 are approached only as an operator becomes singular; gradient ascent reaches 0.924 and 0.796 of the bound at $\delta=0.5$, and finds no violation

of item 6 for $\delta \le 0.2$. Item 25 is resolved for $\delta \ge 0.3$; the search reaches m_{25} to 10^{-3} for $\delta \ge 0.7$ and does not exceed it.

(vi) *Direct sums.* On $M_2 \oplus M_3$ with $W = 1 \oplus s1$ and $W = \mathrm{diag}(1, \ell) \oplus s1$, $s \in \{\frac{1}{4}, 4\}$, intra-block contrast $\delta \in \{0.3, 0.5, 0.7\}$: for central W the largest value found for the items that hold for every trace is 1×10^{-16}; for 11 and 24 the search reproduces the exact values for the fixed density to 2×10^{-11}; with intra-block contrast no value exceeds the M_2 envelope (largest relative excess 1×10^{-16}), and where Theorem 7.1 gives equality the search reaches it to 10^{-3} in 63 of 72 cases.

10 Open problems

Conjecture 10.1. $V_n = \kappa^2 - 1$ for item 7, $V_n = m(\delta)$ for item 10 and $V_n = m_{25}(\delta)$ for item 25 in every dimension; $V_2 = \kappa - 1$ for item 6; and the value of Proposition 4.3 holds for item 9 in every dimension.

Further questions: (1) the envelope of item 6 for $n \ge 3$, where the search exceeds $\kappa - 1$, and whether it grows with n; (2) closed forms for items 12–16, whose leading coefficients the search gives as $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{36}$, 0.064 and 0.09 (item 14: the leading coefficient $(1 - \frac{2}{p})^2/4$ is a lower bound); (3) the analogues for infinite-dimensional von Neumann algebras, where the contrast of an unbounded density may be 1 while the functional is still close to a trace in other senses; (4) envelopes for other inequalities that characterise traces, e.g. those of [25, 15] with the arithmetic mean, and for the dual problem of characterising central elements [26].

Code availability. The library, the program that computes all numbers, tables and the figure, and the tests are available at `https://github.com/andre-a-no/von_neumann_type_i` (MIT licence).

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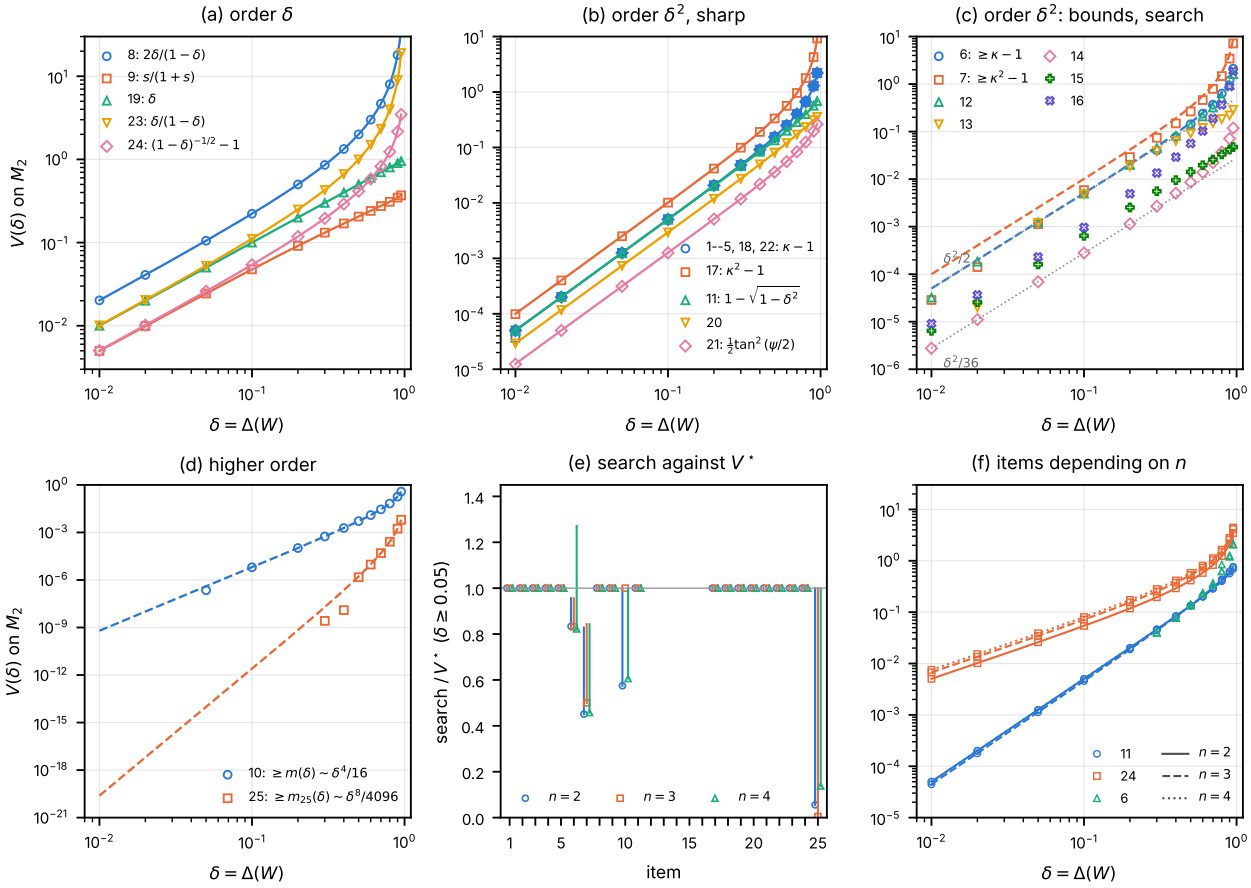


Figure 3: Certified search values on M_2 (symbols) and closed forms or lower bounds (solid and dashed lines), on logarithmic axes, grouped by the order of vanishing: (a) order δ; (b) order δ^2 with sharp values; (c) order δ^2, lower bounds (6, 7) and items without a closed form (12–16; dotted: $\delta^2/2$ and $\delta^2/36$); (d) higher order (10: δ^4; 25: δ^8; the search resolves item 25 only for $\delta \geq 0.3$). (e) The search divided by $V^\star$ for each item and $n = 2, 3, 4$ (segments: range over $\delta \geq 0.05$; symbols: the minimum). Items 6 and 7 reach their suprema only as an operator becomes singular, and the search stays below them; the item-6 values above 1 are certified violations beyond $\kappa - 1$ for $n \geq 3$. (f) The items that depend on n: 11 and 24 (exact, lines) and 6 (search).